Figure 1. CLI crashes on malformed gradient input

Entering 300 without the required colon causes a ValueError when the program tries to unpack drop and distance.

```
E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2020/10/29
Pass Expiry: 2020-10-29 (EXPIRED)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 300
Traceback (most recent call last):
  File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 40, in <module>
    main()
    ~~~~^^
  File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 33, in main
    print("Gradient %:", compute_trail_gradient(e_input))
                         ~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 10, in compute_trail_gradient
    drop_str, dist_str = elevation_str.split(':')
    ^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>
```

Figure 2. Three property tests fail

The initial pytest run reports failures in season pass validity, trail gradient, and lift access tier. The visible traceback begins with the date parser.

```
E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>python -m pytest test_processor.py -v
============================================================================== test session starts ==============================================================================
platform win32 -- Python 3.13.17, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\User\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7
plugins: hypothesis-6.170.15, platformdirs-4.14.5
collected 3 items

test_processor.py::test_parse_season_pass_validity_no_crash FAILED                                                                                                      [ 33%]
test_processor.py::test_compute_trail_gradient_safe FAILED                                                                                                              [ 66%]
test_processor.py::test_calculate_lift_access_tier_safe FAILED                                                                                                          [100%]

================================================================================== FAILURES ==================================================================================
_____________________________________________________________________ test_parse_season_pass_validity_no_crash _____________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 13, in test_parse_season_pass_validity_no_crash
  |     @given(st.text() | st.none())
  |                        ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 16, in test_parse_season_pass_validity_no_crash
    |     result = parse_season_pass_validity(s)
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 3, in parse_season_pass_validity
    |     year = int(parts[0])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 19, in test_parse_season_pass_validity_no_crash
    |     assert False, (
    |     ...<2 lines>...
    |     )
    | AssertionError: parse_season_pass_validity crashed for input '': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_season_pass_validity_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 16, in test_parse_season_pass_validity_no_crash
    |     result = parse_season_pass_validity(s)
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 2, in parse_season_pass_validity
    |     parts = date_str.split('/')
    |             ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 19, in test_parse_season_pass_validity_no_crash
    |     assert False, (
    |     ...<2 lines>...
    |     )
    | AssertionError: parse_season_pass_validity crashed for input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_season_pass_validity_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

Figure 3. Hypothesis exposes gradient parsing errors

The traceback shows that an empty field in the gradient input raises a ValueError during conversion to float.

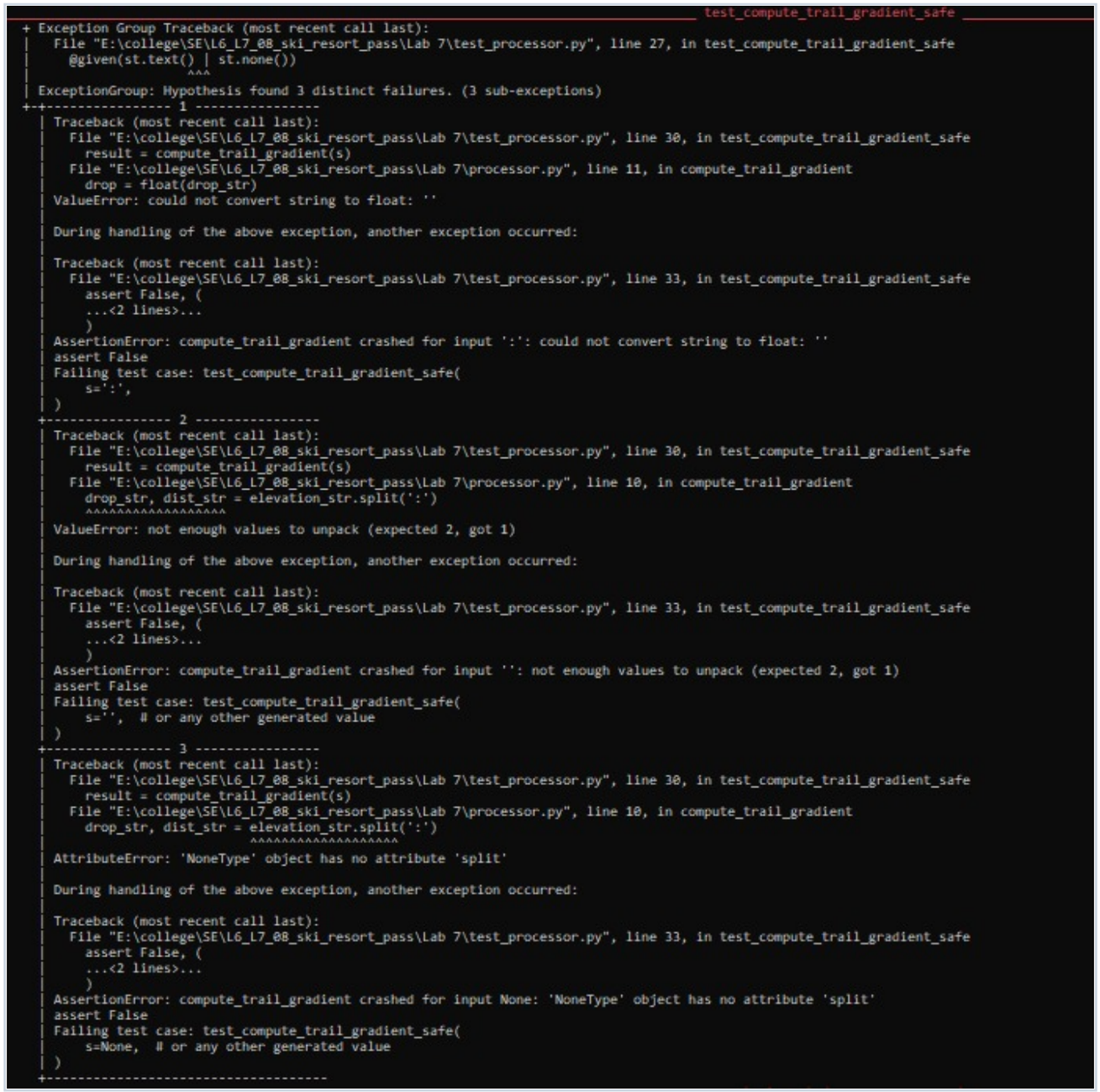


Figure 4. Gradient traceback (duplicate upload)

This is an identical copy of Figure 3, included to account for every uploaded screenshot.

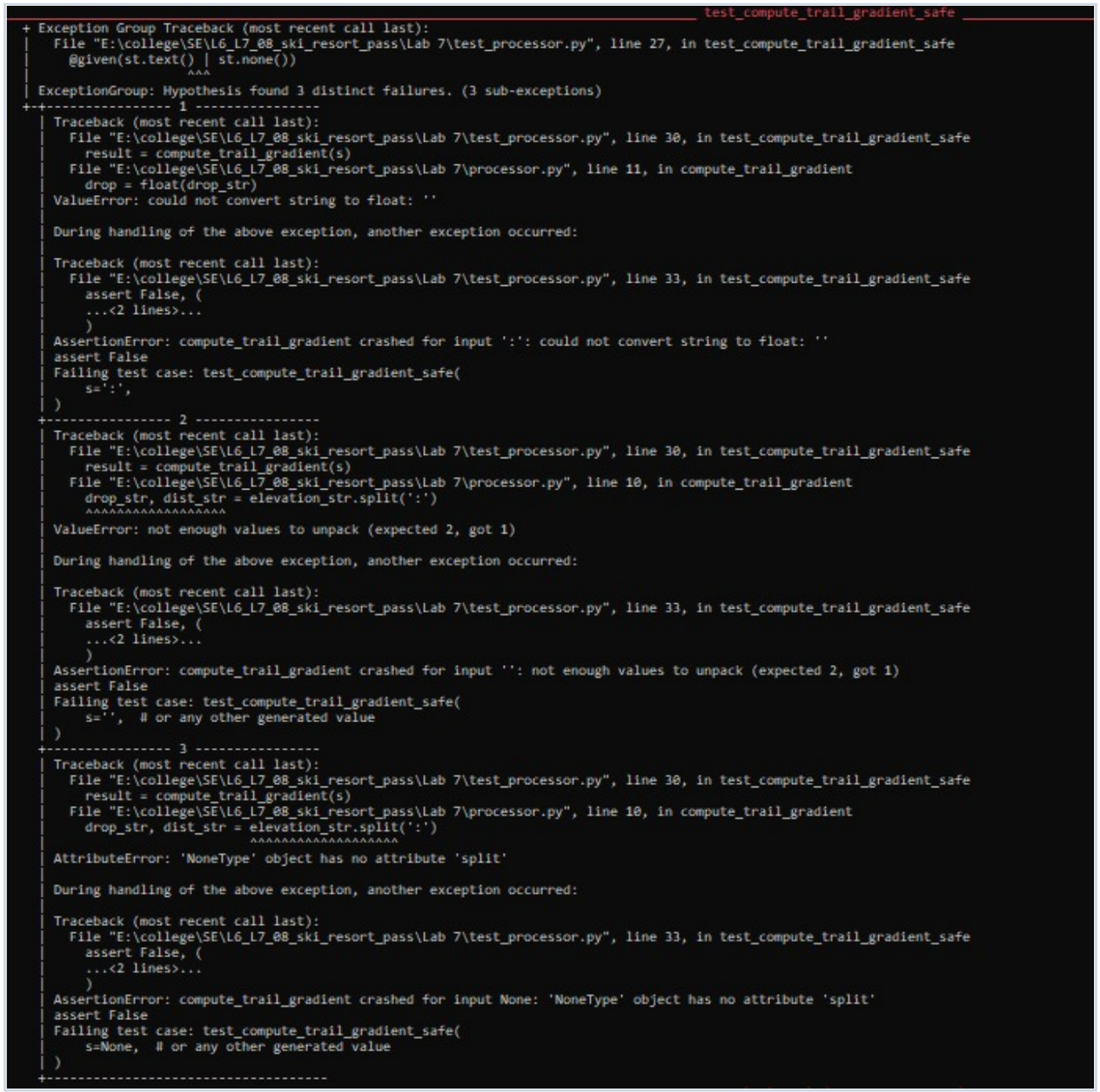


Figure 5. Unknown lift zone raises a KeyError

The original lift-tier function looks up an empty zone prefix directly in its rate table and raises a KeyError.

```
+ Exception Group Traceback (most recent call last):
|   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 41, in test_calculate_lift_access_tier_safe
|     @given(st.text() | st.none())
|                        ^^^
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 44, in test_calculate_lift_access_tier_safe
  |     result = calculate_lift_access_tier(s)
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 24, in calculate_lift_access_tier
  |     return zone_rates[prefix]
  |            ~~~~~~~~~~^^^^^^^^
  | KeyError: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 47, in test_calculate_lift_access_tier_safe
  |     assert False, (
  |     ...<2 lines>...
  |     )
  | AssertionError: calculate_lift_access_tier crashed for input '': ''
  | assert False
  | Failing test case: test_calculate_lift_access_tier_safe(
  |     s='',
  | )
  | Explanation:
  |     These lines were always and only run by failing test cases:
  |         E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py:46
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 44, in test_calculate_lift_access_tier_safe
  |     result = calculate_lift_access_tier(s)
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 17, in calculate_lift_access_tier
  |     prefix = zone_code.strip().upper()[:3]
  |              ^^^^^^^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'strip'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 47, in test_calculate_lift_access_tier_safe
  |     assert False, (
  |     ...<2 lines>...
  |     )
  | AssertionError: calculate_lift_access_tier crashed for input None: 'NoneType' object has no attribute 'strip'
  | assert False
  | Failing test case: test_calculate_lift_access_tier_safe(
  |     s=None,
  | )
  | Explanation:
  |     These lines were always and only run by failing test cases:
  |         E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py:46
  +------------------------------------
=========================== short test summary info ===========================
FAILED test_processor.py::test_parse_season_pass_validity_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_trail_gradient_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_lift_access_tier_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
```

Figure 6. The CLI completes a valid sample run

The sample run marks the 2027/12/25 pass VALID, calculates a 25.0% gradient for 250:1000, and reports $85.00 for the ADV lift tier.

```
E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2027/12/25
Pass Expiry: 2027-12-25 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): ADV
Lift Pass Price: $ 85.0

E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>
```

Figure 7. All three property tests pass

The final pytest run reports three passing tests covering season pass validity, trail gradient, and lift access tier.

```
E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7>python -m pytest test_processor.py -v
============================================================================== test session starts ==============================
platform win32 -- Python 3.13.17, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\User\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: E:\college\SE\L6_L7_08_ski_resort_pass\Lab 7
plugins: hypothesis-6.170.15, platformdirs-4.14.5
collected 3 items

test_processor.py::test_parse_season_pass_validity_no_crash PASSED
test_processor.py::test_compute_trail_gradient_safe PASSED
test_processor.py::test_calculate_lift_access_tier_safe PASSED

============================================================================== 3 passed in 1.20s ================================
```